

ASSIGNMENT 4

Objectives: Sentiment Analysis

- Convert text into numerical features using Bag-of-Words (CountVectorizer), TF-IDF (TfidfVectorizer), and modern vectorization tools.
 - Explain, in your own words, the difference between Bag-of-Words and TF-IDF representations. Which one do you expect to perform better for sentiment analysis, and why?
- Generate a word cloud for positive reviews and negative reviews. Comment on the most frequent words observed in each class.
- Train and evaluate a Naive Bayes classifier and a Logistic Regression classifier for sentiment classification.
- Compare the two models using standard evaluation metrics. Build a comparison table summarizing Accuracy, Precision, Recall, F1-score, and training time for:
 - Naive Bayes + Bag-of-Words
 - Naive Bayes + TF-IDF
 - Logistic Regression + TF-IDF

Based on your results, answer the following:

- (a) Which model performed better overall?
- (b) Which model is more interpretable, and why?
- (c) Which model would you deploy in a production system? Justify your answer.

Use current versions of scikit-learn, pandas, nltk/spaCy, and matplotlib/seaborn.